

Title	Key Ideas	Short Summary(200- 300 words)	Limitations of the work	Suggestions
A Comprehensive Survey on Graph Neural Networks	The paper reviews GNN research and divides GNNs into RecGNN, ConvGNN, GAE, and STGNN. It also provides applications and probable research directions.	Graph Neural Networks are used to work on non-Euclidean data. The paper reviews existing research in GNNs and proposes a new taxonomy which divides GNNs into 4 groups: RecGNN, ConvGNN, GAE, and STGNN, and delves into the techniques used to implement them (primarily supervised and unsupervised learning techniques). The mathematical definitions of all the types are provided for comprehensive understanding (we can use the math to analyze the computational complexities and limitations). Note that ConvGNNs are discussed in high breadth and depth (ConvGNNs are very prevalent in industrial applications). STGNNs can be used to train relativistic data (like high-energy particle physics datasets) . However, we plan to use GNNs for training on multi-omics data (which should primarily use ConvGNNs) [ConvGNNs are also computationally efficient, making them perfect candidates for noisy, heterogeneous (dealing with this is a challenge) multimodal data] .	This is a state of the art baseline paper. Since the paper only discusses possible methods for working on Non-Euclidean Data(like the multi omics data we will work on), it is best to look at the limitations of research papers which align more specifically with our area of interest.	
Using Machine Learning Approaches for Multi-Omics Data Analysis	The paper gives introduction to multi-omics and its subtypes like genomics, transcriptomics, and proteomics; along with techniques used for training on such data.	This paper enunciates the fact that multi-omics data is heterogeneous. There are three integrative methods to train on multi-omics data:- Concatenation-based, Model-based, transformation-based. Concatenation-based method just concatenates different multi-omics data into a single matrix. It requires no preprocessing, but has high dimensionality, noisy data, and risk of overfitting. It is also computationally expensive. Model-based methods use different intermediate models for each data and then combine the results as a combined model for downstream tasks. This approach facilitates the understanding of interactions among different omics for a certain phenotype. Transformation-based approach transforms the data to a uniform mathematical representation and then combines them all before constructing a single model. This approach reduces feature inflation and is great for handling heterogeneous data, but relies on heavy pre-processing and is computationally expensive.	Concatenation-based models are computationally expensive and have very high dimensionality, along with exponential feature explosion. Model-based approach does not help in the understanding of biological interactions as much as the others(as each data is first run through individual models). Transformative approach requires heavy pre-processing and is highly sensitive to hyperparameters(which are manipulated by users, or are automated). Changes in tuning can yield highly variable results, reducing the reliability of the model. This approach also normalizes all data, which can lead to loss of outlier data.	
Deep Multimodal Learning with Missing Modality: A Survey	The paper provides various techniques to handle missing modalities in datasets.Major ways to deal with missing modalities are classified in two major types:- Data Processing Aspect and Strategy Design Aspect	The survey divides the techniques to handle missing modalities into two broad groups:- Data Processing Aspect(recovers missing information) and Strategy Design Aspect(operates without missing data).Major techniques of Data Processing are:- Modality Composition(replaces missing inputs with zero/random values, past data frames (Frame-Repeat), or uses algorithms like KNN to copy data from similar instances of the same category), Modality Generation(uses deep generative architectures like Autoencoders, Generative Adversarial Networks (GANs), or Diffusion Models to synthetically generate the missing modality based on the surviving modalities), Coordinated Representation(Imposes mathematical constraints like Tensor Rank Regularization or Deep Canonical Correlation Analysis during training to align available modalities into a semantically consistent space), Representation Composition(merges surviving modality feature vectors using simple arithmetic operations[max pooling, average pooling] to create a fixed-dimension representation), Representation Generation(trains lightweight feature generators directly or uses auxiliary reconstruction encoder-decoder task to "hallucinate" the latent vector representation of the missing modality without generating raw signals/pixels). Major techniques of Strategy Design are:- Attention-based fusion(this is used in the next paper), Knowledge Distillation (transfers multi-modal insights from a complete-modality "teacher model" down into a "student model" optimized to handle missing elements via response, intermediate, or self-distillation), Graph Learning(using GNNs), MLLMs(using pre-trained foundation networks as feature processors capable of naturally handling absurdly high counts of tokenized data inputs), Ensemble Methods (Combines predictions from multiple distinct unimodal or sub-multimodal expert models), Dedicated Training and Discrete Schedulers(Configue an LLM to act as an agent that interprets input instructions, dynamically routes tasks, and schedules active specialized downstream tools to bypass inactive lines) [Server API protocols can be used for this] .	Efficiency of the methods is not explored in detail (methods should have been compared for computation complexity and favourable use cases). Catching missingness patterns is also underexplored(catching missingness patterns can drastically reduce computation time if we can apply optimized techniques[learnable] to handle them).	The main thing to work on is separating what the paper actually says from your own conclusions. A few of your stronger claims, like STGNNs suiting particle-physics data, or ConvGNNs being "perfect" for multi-omics, are good thinking on your part, but they're written as if the paper proved them. Marking these as your own ideas, the way Gunjan highlights her interpretations, would help coming up with your own hypothesis. And please try to be more precise
MAGNET – Missing-Modality-Aware Graph Neural Network for Cancer Classification	The paper proposes MAGNET which utilizes GNNs capable of handling missing modality for cancer classification. This is accomplished by using Masked Modality matrix and Multi-Head Attention embeddings.	The paper first goes through the challenges of handling missing modality data:- For M modalities, the number of missing patterns is $(2^M)-1$, and artificially introducing missing modalities can oversimplify or misrepresent real-world missingness patterns. MAGNET first uses modality-specific encoders to transform each modality to an embedding(has fixed embedding dimension d). The embedding is converted to a learnable weight matrix using linear transformation on a tensor. Using this, an Attention Coefficient Matrix is calculated. Then a fused embedding matrix is created. Multiple attention heads focus on distinct aspects of the modalities(done by dividing the fused embedding between n attention heads). Then we run the fused embedding through a GraphSAGE for learning. This framework enables us to salvage the usable modalities from a missing-modality dataset . The paper also conducted some experiments comparing MAGNET to top-performing baselines like M3care and MUSE, and documented the findings. The results show that MAGNET scales efficiently with the number of modalities. MAGNET's PMMHA performs better than GAT.	Here the multi-modalities have been defined as the different omics data. In real world examples, multi-modal has a broader definition encompassing text, image, audio, video. We can improve upon existing frameworks by converting, for example, image data into an embedding of the same value range and dimensions as the DNA, mRNA embeddings(this will help seamless integration of varied types of data) .	

Multimodal Patient Representation Learning With Missing Modalities and Labels(MUSE)	The paper proposes MUSE to train on data which has both missing modalities and missing labels. MUSE does it by building a patient-modality bipartite graph to extract modality-agnostic and label-decisive features using mutual-consistent graph contrastive learning.	The paper identifies the challenges of multimodal learning with missing modalities and labels. Problems: A) Missing Modalities and Labels: Existing approaches like Modality Imputation fail to fully use inter-patient relationships. Direct prediction can risk oversimplification of patient-modality relationship or suffer from scalability issues. Labels can also be missing, and existing methods cannot use unlabeled data. B) Modality Collapse: Multimodal models over-rely on a subset of modalities, so they may not generalize well to samples missing key modalities(the models are possibly overfitted) they depended on. To solve these problems, MUSE incorporates: A) Bipartite Patient-Modality Graph: Connects patient nodes directly to available modality nodes, capturing complex data missingness patterns. B) Mutual-Consistent Contrastive Loss: Aligns representations of the same patient across different available data streams. This enables learning modality-agnostic features from unlabeled data. The paper also provides insights: A) Model should accommodate different missingness patterns and effectively model patient-modal relationships. B) Model needs to focus on features that are general across modalities(helps identify missing labels), so that labels can be inferred from existing modalities. The model transforms patient features and modality missingness matrix M to an undirected bipartite patient-modality graph. An edge is drawn if patient p has data from modality m. Node embeddings are initialized using One-Hot encodings for modalities and constant 1 vectors for patients. The model uses multi-layer GNN with Aggregation for learning, applying edge dropout for graph augmentation. The Model also introduces three loss functions: A) Mutual-Consistent Contrastive Loss, B) A second loss function encouraging patient nodes with the same label to predict each other, C) Classification Loss to optimize for downstream tasks. Benchmarking shows the model outperforms baselines like M3Care and GRAPE.	The Model uses a multi-layer GNN which scans the entire graph, which increases computational burden (a GNN with lower computation complexity can be used, but there can be trade-offs like reduced inference of patient-modality relationships in complex systems[this is plausibly because the MUSE GNN looks at information of entire graph, but GNNs like GraphSAGE(which is used in MAGNET) only looks at nearest neighbours]).	Bold text in Red is for personal comments, ideas and proposals. These are unproven inferences drawn from the papers and other sources which may not hold true.
MOGONET integrates multi-omics data using graph convolutional networks allowing patient classification and biomarker identification	The paper proposes MOGONET, a supervised learning based patient classifier and biomarker identifier. It uses individual GCN for each omics data.	The MOGONET model includes three modules: Omics-Specific GCN, View Correlation Discovery Network (VCDN), and alternating joint training. Patient classification and biomarker identification are carried out on the combined prediction level. Omics-Specific GCN: A sample similarity graph is constructed for each omics data type separately, treating modalities like mRNA, DNA methylation, and miRNA as separate views. Then an omics-specific GCN is trained on each graph to produce an initial prediction. Individual prediction embeddings of each omics data should map the specific patient similarities. View Correlation Discovery Network: VCDN is used to combine the initial predictions. The cross-omics relationships of all omics data are explored and fused at the prediction level via late-fusion. Alternating Training: The model utilizes an alternating training process where both networks are trained jointly until convergence. Patient Classification: Multimodal classification tasks, such as BRCA PAM50 subtype classification, are carried out. The mechanism is a supervised learning manner for complete multi-omics data. During testing and optimization, the model learns label-space correlations across the views. In Benchmarking against baselines, it outperformed state-of-the-art multi-omics integrative approaches. However, the original framework lacks a built-in mechanism for data missingness, meaning extra imputation like zero or KNN imputation must be introduced to reconstruct missing modalities.	It uses a GCN which scans the entire graph like MUSE, which is computationally expensive. It is also not designed with missing modalities in mind.	Text in Black is for direct facts and statements proven by papers.
MRGCN: cancer subtyping with multi-reconstruction graph convolutional network using full and partial multi-omics dataset	The paper proposes MRGCN for cancer subtyping using GCN and full/partial multi-omics datasets.	The MRGCN model includes four modules: Individual Encoder, consensus representation, node attribute reconstruction, and graph structures reconstruction. Cancer subtyping is carried out on the consensus representation via spectral clustering algorithm. Individual Encoder: A similarity graph is constructed for each omics data. Then a non-linear encoder is used to create the individual embedding. Individual embedding of each omics data should have same dimension. Consensus Representation: Indicator matrix is created to demonstrate the data missing situation. The consensus representation of all omics data is created using weighted fusion. Node Attribute Reconstruction: The consensus representation preserves main information in multi-omics measurements and similarity relationships in patients. In decoding, the node attributes and graph structures need to be reconstructed simultaneously. The reconstruction is done using a GCN and a Loss function. Another loss function is introduced for graph reconstruction. Cancer Subtyping: Spectral clustering method is used. The mechanism is self-supervised. During training, pseudo labels are generated based on attributes of data and clustering algorithm. Then the model is trained via supervised learning manner by using these as supervised information. Finally, pseudo labels are updated and model is retrained for fine tuning. Kullback–Leibler divergence is used for the learning loss function. For spectral clustering, a similarity matrix is constructed from consensus representation, and results are determined by solving the optimization problem mentioned in the paper. In Benchmarking against baselines, it outperformed algorithms like NEMO and iClusterBayes.	The model use multiple loss functions simultaneously, making hyperparameter tuning time-consuming. The model also creates a separate indicator matrix for demonstrating missing data situations, there could be better ways for doing this(MAGNET's approach is way better [both in terms of computational complexity and capturing relations]).	

M3Care: Learning with Missing Modalities in Multimodal Healthcare Data	The paper proposes M3Care to compensate for the missing information of the patients with missing modalities and perform clinical prediction.	The model includes 4 components: Unimodal Representation Extraction module, Similar Patients Discovery and Information Aggregation module, Adaptive Modality Imputation module, Multimodal Interaction Capture module. Unimodal Representation Extraction: Models like ResNet(for image modalities), MLP(vector based modalities) are used to create latent representation of high dimensionality patient data. Similar Patients Discovery and Information Aggregation: Creates a batch of latent representations of patient data. Then it uses RBF Kernel for calculating similarity between patient data(this is done for modality imputation). But this kernel only works for individual modalities. For multiple modalities, the paper introduces a new, extended kernel. Furthermore, the similarity from each modality is considered to create a filtered similarity matrix. Then the patient batch is represented as a graph, with filtered similarity matrix as graph adjacency matrix. On this, GCN layers are applied to enhance representation learning. Adaptive Modality Imputation: Here, attention fusion is used to adaptively extract the proper amount of information from the previous step. Two weights are used to better connect the representations. Multimodal Interaction Capture: A context-aware multimodal interaction capture is built. It adds internal positional encoding for specifically sequential modalities. Next, representations are added with corresponding modality type embeddings and concatenated to form the input. A cross-entropy loss is used as the prediction loss function. Benchmarking shows that it outperforms baselines like MFN and HGME.	The model uses concatenation for combining modality type embeddings.Simple concanation also has poor scalability for embedding dimensions as more patients are added. Concatenation may be effective here, but methods like individual model training before concatenation(or MAGNET's approach of converting to lower dimension embeddings may better infer patient-modality and patient-patient relationships).
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